where $v = \frac{dx}{dt}$ is the velocity of the loop. This emf is known as motional emf since it is produced due to the movement of the loop in the magnetic field.

From Lenz's law, it is found that the induced current flows in clockwise direction. If R is the resistance of the loop, then the induced current is given by

$$i = \frac{\varepsilon}{R} = \frac{Blv}{R} \qquad \text{(A4.3)}$$

A4.2 ANALOGIES BETWEEN LC OSCILLATIONS AND SIMPLE HARMONIC OSCILLATIONS

Quantitative treatment

The mechanical energy of the spring-mass system is given by

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \qquad \text{(A4.4)}$$

The energy E remains constant for varying values of x and v. Differentiating E with respect to time, we get

$$\frac{dE}{dt} = \frac{1}{2}m\left(2v\frac{dv}{dt}\right) + \frac{1}{2}k\left(2x\frac{dx}{dt}\right) = 0$$

$$\text{or } m\frac{d^2x}{dt^2} + kx = 0 \qquad \text{(A4.5)}$$

$$\text{since } \frac{dx}{dt} = v \text{ and } \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

This is the differential equation of the oscillations of the spring-mass system. The general solution of equation (4.68) is of the form

$$x(t) = X_m \cos(\omega t + \phi) \qquad \text{(A4.6)}$$

where X_m is the maximum value of $x(t)$, ω the angular frequency and ϕ the phase constant.

Similarly, the electromagnetic energy of the LC system is given by

$$U = \frac{1}{2}Li^2 + \frac{1}{2}\left(\frac{1}{C}\right)q^2 = \text{constant} \qquad \text{(A4.7)}$$

Differentiating U with respect to time, we get

$$\frac{dU}{dt} = \frac{1}{2}L\left(2i\frac{di}{dt}\right) + \frac{1}{2C}\left(2q\frac{dq}{dt}\right) = 0$$

$$\text{or } L\frac{d^2q}{dt^2} + \frac{1}{C}q = 0 \qquad \text{(A4.8)}$$

$$\text{since } i = \frac{dq}{dt} \text{ and } \frac{di}{dt} = \frac{d^2q}{dt^2}$$

The general solution of equation (A4.8) is of the form

$$q(t) = Q_m \cos(\omega t + \phi) \qquad \text{(A4.9)}$$

where Q_m is the maximum value of $q(t)$, ω the angular frequency and ϕ the phase constant.

A5.1 PROPERTIES OF ELECTROMAGNETIC WAVES

- The energy density (energy per unit volume) associated with an electromagnetic wave propagating in vacuum or free space is

$$u = \frac{1}{2}\varepsilon_\circ E^2 + \frac{1}{2\mu_\circ}B^2$$

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